SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

FOR

Student Management System (SMS) – Registration Module Initial version

Version 1.0

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Preface

This document contains the Software Requirements Specification (SRS) of a Student Management System for the Department of software engineering at the General Sir John Kotelawala Defence University. The main aim of this project is to add functionality to the SMS system to provide an online facility for managing and registering student accounts.

This document has been prepared following the IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications [IEEE 1998].

1. Introduction

This Software Requirement Specification is written in accordance with the IEEE Std 830-1998 model.

1.1. Purpose

This SRS Document contains the complete software requirements for the Student Management System and describes the design decisions, architectural design and detailed design needed to implement the system. It provides the design visibility and information needed for software support.

1.2. Scope

A student Management System will be developed for the Department of software engineering at the General Sir John Kotelawala Defence University and used to replace the old paperwork system. The student Management System should be a web-based project deployed alone with the existing systems in the cloud environment. This increase in the efficiency of student registrations and audit trails of internal processes. It provides a mechanism to edit student details, which makes the system flexible.

1.3. Overview

This document has been prepared in accordance with the IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications [IEEE 830-1998 (1998)]. It provides information on Product perspective, Product functions, User characteristics, Constraints, Assumptions and dependencies and specific requirements.

2. Overall description

This section of the SRS describes all general factors of the product and its requirements.

2.1. Product perspective

2.1.1. System interfaces

Student Management System (SMS) web-based system will be developed as microservices. This system can connect to relational databases. The system can also be containerized and support container orchestration environments.

2.1.2. User interfaces

All pages of the system follow a consistent theme and clear structure. The occurrence of errors should be minimized through validation for input from the user. The error message should be located beside the error input, clearly highlighting and telling users how to solve it.

2.1.3. Hardware interfaces

a. Server Side

The web application will be hosted on one of the Department's Linux servers and connected to one Database server.

b. Client Side

The system is a web-based application; clients require a modern web browser such as Mozilla Firefox, Internet Explorer and Google Chrome. The computer must have an Internet connection to be able to access the system.

2.1.4. Software interfaces

a. Server Side

The university provides all the Linux-based servers, and this software can be deployed on them.

b. Client Side

An OS is capable of running a modern web browser that supports HTML. Clients also use the mobile browser for using this software.

2.1.5. Communications interfaces

The HTTP protocol will facilitate communications between the client and server.

2.2. System functions

This section outlines all the main features of the Student Management System (SMS).

2.2.1. Administration role

The system administrator must be able to:

1. Add the students
2. Remove the students
3. List the students
4. Alter the student's course details

2.2.2. Audit Trailing

All the administration activities need to be appropriately logged in the system.

3. Specific requirements

3.1. Functional requirements

3.1.1. User class – Administrator

1. The student registration form consists of student details such as the student's first name, last name, student address, student birthday, student ID number, student degree program, and student enrolled courses.
2. Based on the registered student will get a unique student number.
3. Every semester administrator changes the student courses.
4. Student course history should need to be maintained throughout the degree.
5. At any time, the administrator can search for the student from the unique ID and get all the details.

3.2 Design constraints

The system needs to design using the latest tools and technologies.

3.3 Software system attributes

3.3.1 Security

The system must log the client's every activity properly for future reference.

3.3.2 Maintainability

The system should need support for the database credentials changes with minimal impact.

3.3.3 Portability

It should be transferable between different OS.